

Topic 12: Undetermined Coefficients & The Annihilator Method

Professor Leonard Master Series · Undetermined Coefficients

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: The Educated Guess

When solving non-homogeneous equations:

$$ay'' + by' + cy = g(x)$$

The general solution is:

$$y(x) = y_c(x) + y_p(x)$$

- $y_c(x)$ is the complementary solution to $ay'' + by' + cy = 0$.
- $y_p(x)$ is a particular solution that produces $g(x)$.

The **Method of Undetermined Coefficients** is based on a simple observation: if $g(x)$ is a polynomial, exponential, or sine/cosine, its derivatives look like variations of the same family. So we make an **educated guess** with unknown coefficients A, B, C , plug it into the left-hand side, and solve for those coefficients!

2. Leonard's Master Guessing Table

Driving Term $g(x)$	Trial Particular Solution $Y_p(x)$
k (constant)	A
$3x + 5$	$Ax + B$
$x^2 - 1$	$Ax^2 + Bx + C$
e^{5x}	Ae^{5x}
$\sin(3x)$ or $\cos(3x)$	$A \cos(3x) + B \sin(3x)$ (Must include BOTH!)
xe^{2x}	$(Ax + B)e^{2x}$
$e^x \cos(2x)$	$e^x (A \cos(2x) + B \sin(2x))$

3. The Number One Trap: The Duplication Rule

"If any term in your guess Y_p already appears in y_c , that term will be annihilated to 0 by the left-hand side!"

The Fix: Multiply your guess by x (or x^2 if it was a repeated root) until NO term in Y_p matches any term in y_c .

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Example

: $y'' - 4y' + 4y = e^{2x}$.

- Characteristic eq: $(r - 2)^2 = 0 \implies y_c = C_1e^{2x} + C_2xe^{2x}$.
- Guessing $Y_p = Ae^{2x}$ fails (duplicates C_1e^{2x}).
- Guessing $Y_p = Axe^{2x}$ fails (duplicates C_2xe^{2x}).
- **Correct Guess:** Multiply by x^2 : $Y_p = Ax^2e^{2x}$!